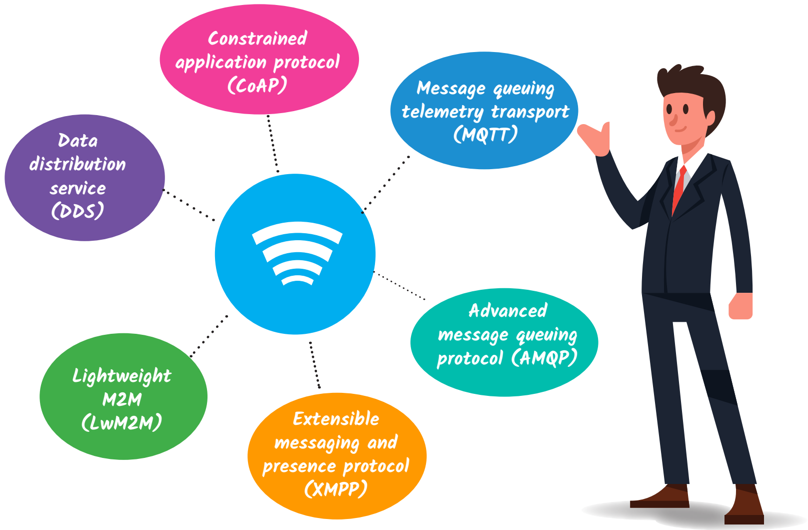
Choose the Most Optimal IoT Protocol for Your Project

The Internet of Things (IoT) is a term used to describe the interconnectivity and intercommunication of everyday devices with the Internet. Everything, from a smartwatch to a smart light bulb, is part of IoT. In this article, we talk about the different protocols used to enable easy communication between IoT devices. We also discuss which IoT protocol might be ideal for your project. If you are an IoT enthusiast looking to get started with your own project, this article is a must read.



It all started in 1982 with a Coca Cola vending machine developed at Carnegie Mellon University. Designed on the back of ARPANet, this machine had the ability to report the number of Coca Cola cans still left in the machine and whether these were cold. This basic idea of combining the Internet with regular day-to-day devices or 'things' took root and resulted in the birth of what is today known as the Internet of Things (IoT). The term IoT itself was coined in 1999 by Kevin Ashton, the co-founder of the Auto-ID center at MIT.

Since then, IoT has come a long way, evolving into a huge industry with millions of active devices serving a broad spectrum of applications. Smart home devices such as Google Home and Amazon Alexa are good examples of how IoT is being deployed by major tech companies to provide a wide range of user-friendly services. From smart TVs and refrigerators to smart surveillance cameras and lights, IoT is everywhere today.

As shown in Figure 1, by 2025, over 75 billion active IoT devices are estimated to function across the globe. With the AI revolution well on its way, IoT plays a key role in the future of electronics. Everything, from self-driving cars to automated train and aeroplane systems, has been made possible with IoT. Major tech companies such as Intel, Dexcom and Cisco have invested heavily into developing IoT devices and circuits. If you are looking to get started with getting your feet wet in the field of IoT, there was never a better time than now.

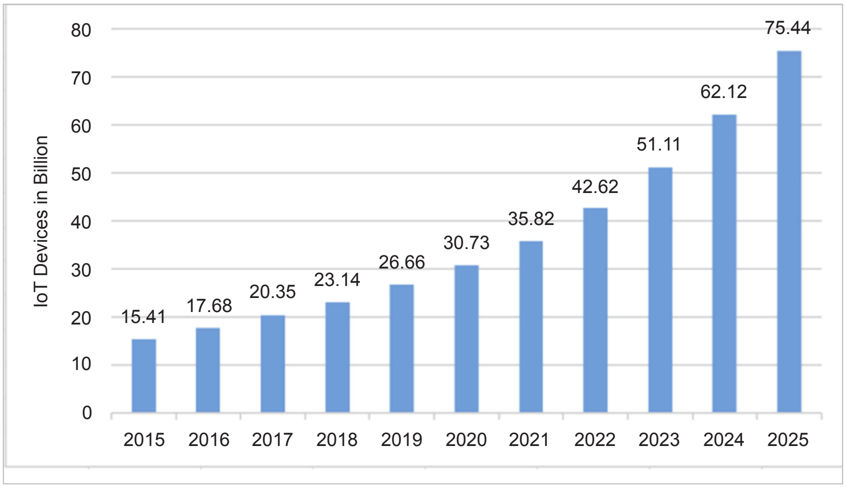


Figure 1: Projections of growth in IoT devices by 2025 (Courtesy: ResearchGate)